INCH-POUND

MS53231B
W/AMENDMENT 1
1 August 2023
SUPERSEDING
MS53231B
24 August 2007

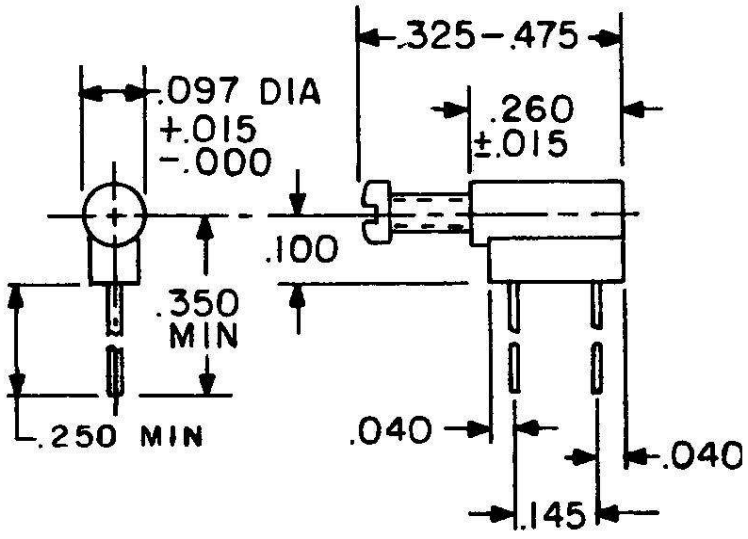
MILITARY SPECIFICATION SHEET

# COILS, RADIO FREQUENCY, ENCAPSULATED, VARIABLE, MICRO-MINIATURE, IRON CORE, TYPES LT11V

Inactive for new design, after 31 March 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.




| INCHES | mm |
|---|---|
| .015 | 0.38 |
| .040 | 1.02 |
| .097 | 2.46 |
| .100 | 2.54 |
| .145 | 3.68 |
| .250 | 6.35 |
| .260 | 6.60 |
| .325 | 8.26 |
| .350 | 8.89 |
| .475 | 12.07 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Dimensions and configuration.

AMSC N/A
FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT11V

Grade: 2
Class: A

Weight: .2 gram, maximum.

Operating temperature range: -55° to +105°C.

Ambient temperature: +90°C maximum.

Temperature rise: +15°C.

Power dissipation: 225 mW maximum.

Lead material: -1 through 12 is .018 (0.46 mm) diameter copper.
-13 through –21 is .018 (0.46 mm) diameter gold-plated Kovar.

Terminal pull: 2 pounds minimum.

Tuning torque: .005 - .15 in. oz.

Stop torque: .2 in. oz. max.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 1,000 V rms minimum.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 300 V rms minimum.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Marking: Marking shall be as specified in MIL-PRF-15305 except that the marking shall be on the unit package or container.

Part or Identifying Number (PIN): MS53231 - (dash number from table I).



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TABLE 1. Electrical characteristics (initial).

| Dash Number MS53231 1/ | Inductance L Max. µH | Inductance L Min. µH | Q Min. At L Max. | Test Frequency (MHz) | SRF Minimum (MHz) | DC resistance At 25°C, max (Ohms) | Rated DC current, maximum (mA) |
|---|---|---|---|---|---|---|---|
| -1 | .025 | .020 | 50 | 200 | 1,000 | .02 | 250 |
| -2 | .051 | .035 | 40 | 100 | 700 | .03 | 250 |
| -3 | .076 | .040 | 40 | 100 | 615 | .10 | 250 |
| -4 | .10 | .07 | 30 | 25 | 500 | .30 | 250 |
| -5 | .15 | .11 | 30 | 25 | 420 | .35 | 200 |
| -6 | .22 | .16 | 30 | 25 | 400 | .40 | 180 |
| -7 | .27 | .19 | 30 | 25 | 400 | .50 | 180 |
| -8 | .33 | .165 | 30 | 25 | 320 | .60 | 140 |
| -9 | .47 | .34 | 30 | 25 | 290 | .65 | 100 |
| -10 | .56 | .40 | 30 | 25 | 250 | .75 | 100 |
| -11 | .68 | .50 | 30 | 25 | 240 | .85 | 100 |
| -12 | .82 | .59 | 30 | 25 | 180 | .90 | 100 |
| -13 | 1.0 | .50 | 25 | 7.9 | 140 | .40 | 125 |
| -14 | 1.5 | 1.0 | 25 | 7.9 | 135 | .80 | 100 |
| -15 | 2.2 | 1.2 | 25 | 7.9 | 100 | 1.4 | 80 |
| -16 | 2.7 | 1.95 | 25 | 7.9 | 60 | 1.6 | 75 |
| -17 | 3.3 | 2.2 | 25 | 7.9 | 60 | 1.8 | 65 |
| -18 | 4.7 | 2.5 | 25 | 7.9 | 50 | 2.2 | 60 |
| -19 | 5.8 | 2.9 | 25 | 7.9 | 38 | 2.8 | 58 |
| -20 | 6.8 | 3.4 | 25 | 7.9 | 36 | 3.0 | 55 |
| -21 | 8.2 | 4.6 | 25 | 7.9 | 25 | 3.0 | 50 |

1/ The dash number added to the MS military standard number constitutes the MS part number, for example MS53231-1



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TABLE II. Electrical characteristics (final).

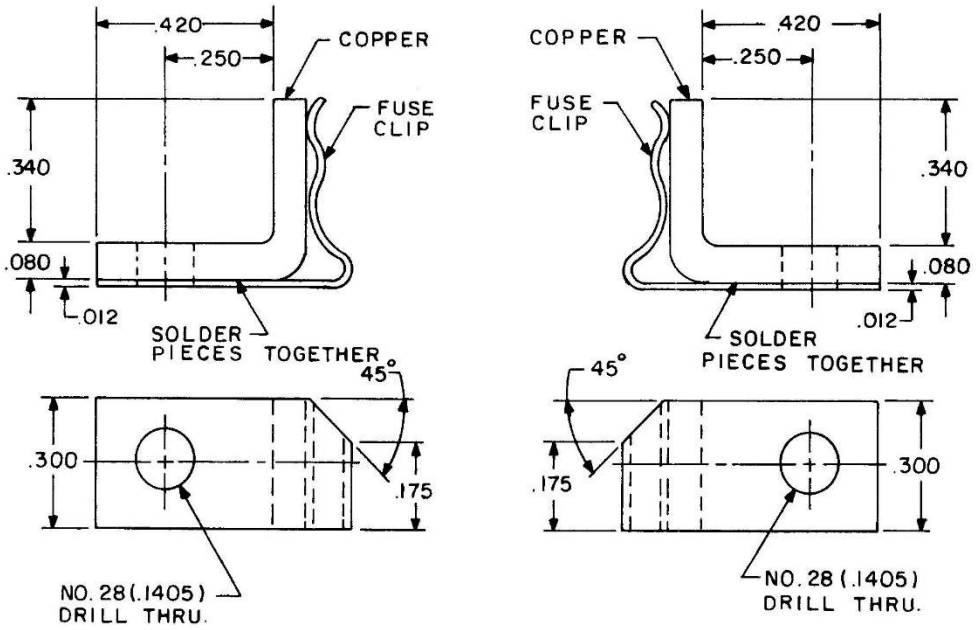
| Inspection group | Allowable variation from Initial measurement | Allowable variation from Initial measurement | Allowable percent from specified minimum value in electrical characteristics (initial) table | Allowable percent from specified minimum value in electrical characteristics (initial) table |
|---|---|---|---|---|
| | Inductance (percent) | DC resistance | Self-resonant frequency | Q |
| Qualification inspection | | | | |
| Group II | ±5 | --- | --- | -10 |
| Group III | ±5 | ±(2% +.001 ohm) | -5 | -10 |
| Group IV | ±5 | ±(2% +.001 ohm) | -10 | -10 |
| Conformance inspection group C | | | | |
| Subgroup I | ±5 | --- | --- | -10 |
| Subgroup II | ±5 | ±(2% +.001 ohm) | -10 | -10 |
| Subgroup III | ±5 | ±(2% +.001 ohm) | -5 | -10 |

Application notes:

1. The test fixture in the diagram following shall be used for electrical measurements. Inductance values are effective inductance as indicated on a HP260A, HP190A or equivalent Q meter, when tested in the test fixture. Add 5 percent to Q reading to account for loss of Q in the test jig.
2. Polarization during moisture resistance test is not applicable.
3. Coils are held rigidly by the body during vibration and mechanical shock testing.
4. Resistance to soldering heat test, per MIL-STD-202-210, test condition D, is applicable.
5. For dielectric withstanding voltage, barometric pressure and insulation resistance units shall be placed on Flat metal plate with leads insulated from surface. Measurement of dielectric withstanding voltage, barometric pressure and insulation resistance shall be between the leads of the coil connected together and the metal plate.
6. Screw core assembly shall be set at maximum specified inductance value indicated in the electrical characteristics table (initial), prior to all inspection tests. This setting shall not be changed until electrical characteristics (final) measurements are performed.



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TEST FIXTURES FOR ELECTRICAL MEASUREMENTS

| | INCHES | MM | |
|---|---|---|---|
| | .012 | 0.30 | |
| SIDE B | .080 | 2.03 | SIDE A |
| | .1405 | 3.570 | |
| | .175 | 4.45 | |
| | .250 | 6.35 | |
| | .300 | 7.62 | |
| | .340 | 8.64 | |
| | .420 | 10.67 | |

FIGURE 2. Test fixture for electrical measurements.



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Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-210 MIL-STD-202-213 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2023-018)

Review activities:
Army – AR, MI
Navy – AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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